ALLOPURINOL

Do not start during acute attack

Severe skin reactions

CHEMICAL NAME 1H-pyrazolo[3,4-d]pyrimidin-4(5H)-one
CLASS Xanthine oxidase inhibitor (antigout / urate-lowering drug)
TRADE NAMES Zyloric · Zyloprim · Allopurinol (generic)

THERAPEUTIC USES / INDICATIONS

- Prevention of recurrent gout and gouty tophi (long-term urate lowering).
- Prevention of tumour lysis syndrome / hyperuricaemia with chemotherapy.
- Uric acid and calcium oxalate kidney stones.

ADMINISTRATION

- **PO** once daily after food with plenty of water.
- Doses >300 mg may be divided.

DOSAGES

- **Gout:** start 100 mg daily (50 mg in renal impairment), increase by 100 mg every 2–4 weeks to target serum urate <360 µmol/L (6 mg/dL); max 900 mg/day.
- **Tumour lysis prevention:** 300–600 mg/day starting 1–2 days before chemotherapy.
- **Children (malignancy):** 10–20 mg/kg/day.
- Give colchicine or NSAID cover for the first 3–6 months.

PHARMACOKINETICS

- **Onset:** urate falls in 2–3 days; full effect 1–3 weeks.
- Metabolised in liver to oxipurinol (active, long-acting).
- Excreted in urine; accumulates in renal impairment. **t½:** 1–2 h (oxipurinol ~15 h).

PHARMACODYNAMICS

- Inhibits **xanthine oxidase** → ↓ conversion of hypoxanthine → xanthine → uric acid.
- Lowers serum and urinary uric acid.
- Initial urate shifts can trigger acute gout flares.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Skin rash (stop drug if rash appears).
- Nausea, diarrhoea.
- Acute gout flare at start of treatment.

ADVERSE EFFECTS (SERIOUS)

- SJS/TEN and DRESS (hypersensitivity syndrome) — higher risk with HLA-B*5801.
- Hepatotoxicity.
- Bone-marrow suppression.
- Interstitial nephritis.

CONTRAINDICATIONS & CAUTIONS

- Previous hypersensitivity reaction to allopurinol.
- Starting during an acute gout attack (continue if already taking).
- **Caution:** renal or hepatic impairment; HLA-B*5801 carriers (Han Chinese, Thai, Korean).

KEY INTERACTIONS

- **Azathioprine, mercaptopurine:** toxicity; reduce their dose to 25%.
- **Amoxicillin/ampicillin:** ↑ rash.
- **Warfarin:** ↑ INR.
- **Thiazides:** ↑ hypersensitivity risk.

NURSING MANAGEMENT

- Confirm acute attack has settled before starting.
- Encourage 2–3 L fluid/day to prevent stones.
- Monitor serum urate, renal function, LFTs and CBC.
- Stop and report any rash immediately.
- Check for azathioprine use.

PATIENT EDUCATION

- Take daily even when well — it prevents attacks, does not treat them.
- Stop and seek help if a rash, fever or mouth sores appear.
- Drink plenty of water; limit alcohol and high-purine foods.
- Flares may occur early; keep taking prescribed colchicine/NSAID.

KEY POINTS

- Xanthine oxidase inhibitor; prevents gout.
- Rash → stop (SJS/DRESS risk).
- Start low (100 mg), go slow; never start in acute attack.
- Reduce azathioprine dose.

SOURCES CHECKED [51] Allopurinol 100 mg Tablet (UK SmPC) [52] Allopurinol, oral (US label) [53] Allopurinol, IV (US label) *Full references p. 432–457*